

NeuralCanvas
Real-Time AI Style Transfer

Intelligent Web Systems (22AI025)

Submitted by

Vasudev Rai (2210993857)

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Guided By

Mr. Yogendra Sharma

**CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING &
TECHNOLOGY CHITKARA UNIVERSITY, RAJPURA**

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With immense pleasure I,

Mr. Vasudev Rai , presenting

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Vasudev Rai

2210993857

ABSTRACT

NeuralCanvas – Real-Time AI Style Transfer is a deep learning-based system designed to generate artistic images by combining the content of one image with the visual style of another. Neural Style Transfer is a technique that uses convolutional neural networks to extract and blend content and style representations from images. In this project, a fast and efficient neural style transfer model is implemented using a lightweight transformer network inspired by MobileNet architecture.

The system utilizes a pre-trained VGG16 network as a loss network to extract high-level feature representations, which are used to compute content loss and style loss during training. Style patterns are captured using Gram matrices, enabling the model to learn textures, colors, and artistic patterns from the style image.

The model is trained on a dataset of natural images and learns to apply a specific artistic style to input images while preserving their structural content. Once trained, the system can perform stylization quickly, enabling real-time artistic transformation of images. Additionally, the project includes a webcam-based module that allows live style transfer on video frames, demonstrating real-time performance. The implementation is developed using Python and the PyTorch deep learning framework. Experimental results show that the proposed system successfully produces visually appealing stylized images while maintaining computational efficiency.

The **NeuralCanvas** system highlights the potential of deep learning techniques in creative image processing, digital art generation, and real-time AI-based visual applications.

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INTRODUCTION

With the rapid advancement of artificial intelligence and deep learning, computers are increasingly capable of performing complex visual tasks such as image recognition, object detection, and artistic image generation. One of the most fascinating applications of deep learning in computer vision is **Neural Style Transfer**, a technique that combines the content of one image with the artistic style of another to generate visually appealing stylized images. This technique enables machines to recreate artistic effects similar to famous painting styles while preserving the structure and content of the original image.

Traditional image editing techniques require manual effort and specialized artistic skills to create stylized images. Neural Style Transfer automates this process by using deep convolutional neural networks to learn representations of both image content and artistic style. By extracting high-level features from images, neural networks can understand the structure of objects in an image as well as the texture patterns that define artistic styles. This makes it possible to generate images that maintain the original content while adopting the style of another image.

The project “**NeuralCanvas – Real-Time AI Style Transfer**” focuses on implementing a fast and efficient neural style transfer system using deep learning techniques. The system utilizes a transformer network inspired by a lightweight MobileNet architecture to perform fast image stylization. In addition, a pre-trained VGG16 convolutional neural network is used as a feature extractor to compute content and style representations during the training process. The style of an image is captured using Gram matrices, which represent the correlation between feature maps and effectively encode artistic patterns such as textures, brush strokes, and color distributions.

A key objective of this project is to achieve **real-time performance** while maintaining high-quality artistic outputs. Once trained, the model can quickly transform input images into stylized versions without requiring complex computations. The system also includes a webcam-based implementation that applies style transfer to live video frames, demonstrating the capability of performing artistic transformations in real time.

The implementation of NeuralCanvas is developed using **Python and the PyTorch deep learning framework**, along with supporting libraries such as Torchvision, OpenCV, and PIL for image processing and data handling. The proposed system demonstrates how deep learning can be applied to creative domains such as digital art generation, visual media editing, and interactive AI applications.

Overall, the NeuralCanvas project highlights the potential of deep learning techniques in combining technology and creativity, enabling machines to produce artistic visual outputs efficiently and in real time.

PROBLEM FORMULATION

Creating artistic images traditionally requires significant manual effort and artistic expertise. Designers and artists often rely on specialized software and editing techniques to apply artistic styles such as paintings, sketches, or abstract effects to photographs. However, these methods can be time-consuming and require professional skills. With the growth of digital media and social platforms, there is an increasing demand for automated tools that can quickly generate artistic visual content.

Neural Style Transfer provides a solution to this problem by using deep learning techniques to automatically combine the **content of one image with the style of another image**. While early neural style transfer methods produced high-quality results, they were computationally expensive and required several iterations of optimization for each image. This made them unsuitable for real-time applications.

The main problem addressed in this project is how to **efficiently generate stylized images in real time while maintaining the visual quality of the original content**. The challenge lies in preserving the structural details of the input image while accurately transferring the textures, colors, and artistic patterns of the style image. Additionally, the system must be computationally efficient so that it can perform stylization quickly, even on systems with limited hardware resources.

The **NeuralCanvas – Real-Time AI Style Transfer** system aims to address these challenges by developing a lightweight neural network model capable of performing fast style transfer. The project uses a transformer network based on a MobileNet-inspired architecture for efficient image transformation, while a pre-trained VGG16 network is used to extract meaningful feature representations for computing style and content losses during training.

By training the model to learn the relationship between content and style features, the system can apply a predefined artistic style to any input image quickly and efficiently. The final solution enables real-time image stylization and supports applications such as creative image editing, artistic photo filters, and live webcam-based style transformation.

METHODOLOGY

The methodology explains the process used to design and implement the neural style transfer system.

1. Data Preparation

- A dataset of natural images is used as **content images** for training.
 - A **style image** is selected that represents the artistic pattern to be transferred.
 - Images are preprocessed before training.
 - Preprocessing includes:
 - Image resizing
 - Center cropping
 - Conversion to tensor format
 - Normalization of pixel values
-

2. Feature Extraction

- Feature extraction is performed using the pre-trained VGG16 model.
 - The model extracts hierarchical features from images.
 - Different layers capture different information:
 - Low-level features → edges and textures
 - Mid-level features → patterns and shapes
 - High-level features → object structure
 - These extracted features are used to calculate **content loss and style loss**.
-

3. Style Representation using Gram Matrix

- The artistic style of an image is represented using the **Gram Matrix**.
 - Gram Matrix calculates the correlation between feature maps.
 - It captures:
 - Texture patterns
 - Brush strokes
 - Color distributions
 - The generated image style is matched with the style image using Gram Matrix comparison.
-

4. Transformer Network Architecture

- The system uses a **Transformer Network based on MobileNet architecture**.
 - The network includes:
 - Convolution layers for feature extraction
 - Bottleneck (residual) blocks for efficient learning
 - Upsampling layers for image reconstruction
 - Instance normalization for training stability
 - The network transforms the input image into a stylized output image.
-

5. Loss Function Computation

The training process uses multiple loss functions:

Content Loss

- Preserves the structure and objects of the input image.

Style Loss

- Transfers artistic style patterns from the style image.

Regularization Loss

- Ensures smoothness and reduces noise in the output image.

Total Loss is calculated as:

- $\text{Total Loss} = \text{Content Loss} + \text{Style Loss} + \text{Regularization Loss}$
-

6. Model Training

- The neural network is trained using the **Adam optimizer**.
 - Model parameters are updated to minimize the total loss.
 - Training is performed for multiple epochs.
 - Intermediate stylized outputs are generated to monitor training progress.
-

7. Image Stylization

- After training, the model can stylize new images.
 - The input image is passed through the trained transformer network.
 - The network generates the stylized image in a **single forward pass**.
 - This makes the system **fast and efficient**.
-

8. Real-Time Webcam Stylization

- The system supports real-time style transfer.
- Webcam frames are captured using **OpenCV**.
- Each frame is passed through the trained neural network.
- The stylized frames are displayed in real time.

FLOWCHART

Workflow of NeuralCanvas – Real-Time AI Style Transfer

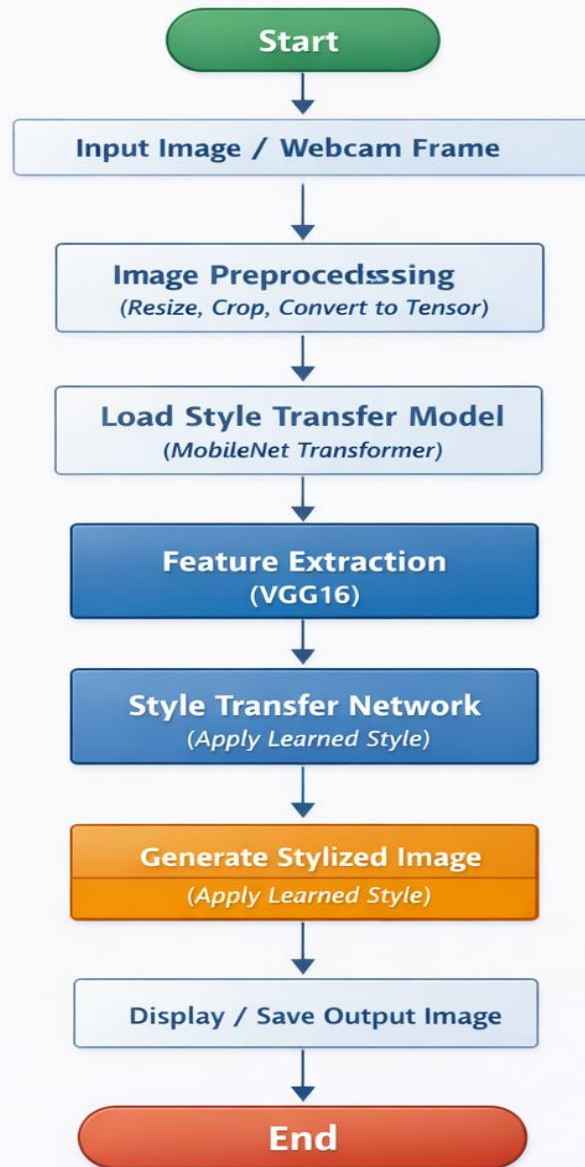


FIG 1

SOFTWARE & HARDWARE REQUIREMENTS

1. Software Requirements

Development Environment:

- **Python:** Primary programming language used for implementing the neural style transfer model.
- **Visual Studio Code / PyCharm:** Code editors used for writing, debugging, and managing the project.
- **Git & GitHub:** Used for version control and project management.

Deep Learning Framework:

- **PyTorch:** Core deep learning framework used to build and train the neural style transfer model.
- **Torchvision:** Provides pre-trained models, datasets, and image transformation utilities.

Image Processing Libraries:

- **OpenCV:** Used for real-time webcam capture and image processing.
- **PIL (Python Imaging Library):** Used for loading, processing, and saving images.
- **NumPy:** Used for numerical operations and tensor manipulation.

Machine Learning Components:

- **VGG16 Pretrained Model:** Used as a feature extractor for computing style and content loss.
- **MobileNet-based Transformer Network:** Used for efficient and fast style transfer.

Supporting Libraries:

- **TQDM:** Used for displaying training progress bars.
- **Argparse:** Used for command-line argument handling.
- **OS Library:** Used for file management and directory handling.

Package Management:

package manager used to install required dependencies.

Operating System:

- Windows 10 / Windows 11
- Linux distributions (Ubuntu, Fedora, etc.)
- macOS

2. Hardware Requirements

For Development and Training:

- **Processor:** Minimum Intel Core i5 or AMD Ryzen equivalent (Quad-Core recommended).
- **RAM:** Minimum 8 GB (16 GB recommended for faster training).
- **Storage:** At least 20 GB of free space (SSD recommended for better performance).
- **Graphics Processing Unit (Optional but Recommended):** NVIDIA GPU with CUDA support for faster deep learning model training.

For Real-Time Execution:

- **Processor:** Multi-core CPU capable of handling image processing tasks.
- **RAM:** Minimum 8 GB.
- **Webcam:** Required for real-time style transfer using live video frames.
- **Display:** Monitor capable of rendering real-time stylized video output.

For End Users:

- **Device:** Laptop, desktop, or workstation capable of running Python applications.
- **Webcam:** Required for real-time style transfer functionality.
- **Internet:** Optional (only required for downloading datasets or models).

• .

EXPERIMENTS

1. Image Style Transfer Generation

Objective:

To verify that the system can successfully transform a content image into an artistic image using a trained neural style transfer model.

Procedure:

Loaded the trained model file (.pth) into the NeuralCanvas system. Provided an input image as the content image and processed it through the transformer network. The system applied the learned artistic style from the style model and generated a stylized output image. Multiple test images were used to evaluate the model's performance.

Result:

The system successfully generated stylized images while preserving the original structure of the input image. The artistic patterns from the style model were applied effectively, producing visually appealing results.

2. Style Model Testing

Objective:

To test whether different style models produce distinct artistic outputs.

Procedure:

Used multiple pretrained style models such as mosaic, candy, and picasso. Each model was applied to the same input image to compare the generated outputs. The results were analyzed to verify that the artistic patterns corresponded to the selected style model.

Result:

Each model generated a unique artistic output corresponding to the trained style. The system successfully applied different artistic styles without affecting the original content structure.

3. Neural Network Training

Objective:

To train the neural style transfer model to learn the artistic features of a style image.

Procedure:

Used a dataset of natural images as content images for training. Selected a style image and extracted style features using the VGG16 feature extraction network. Computed content loss, style loss, and regularization loss during training. The transformer network parameters were updated using the Adam optimizer.

Result:

The model successfully learned the style patterns of the input style image. Training produced a style model file that could be used to stylize new images efficiently.

4. Real-Time Webcam Style Transfer

Objective:

To test the real-time performance of the NeuralCanvas system using a webcam.

Procedure:

Integrated OpenCV to capture frames from the webcam. Each frame was preprocessed and passed through the trained neural network. The stylized output frames were displayed continuously to evaluate real-time processing capability.

Result:

The system successfully applied style transfer to live webcam frames with minimal delay. The stylized video output maintained smooth performance and visual quality.

5. Image Processing and Preprocessing

Objective:

To verify that image preprocessing operations correctly prepare input images for the neural network.

Procedure:

Applied preprocessing steps including image resizing, center cropping, tensor conversion, and normalization. Multiple image sizes and formats were tested to ensure compatibility with the neural network model.

Result:

The preprocessing pipeline successfully prepared images for neural network processing, ensuring consistent input dimensions and improved model performance.

6. Model Performance Evaluation

Objective:

To evaluate the performance and efficiency of the NeuralCanvas system.

Procedure:

Tested the system using multiple input images and measured the time required to generate stylized outputs. Performance was evaluated on CPU-based systems and compared with traditional optimization-based style transfer methods.

Result:

The NeuralCanvas system produced stylized images quickly due to the feed-forward transformer network. The model demonstrated efficient performance suitable for real-time applications.

RESULTS



FIG 2



FIG 3

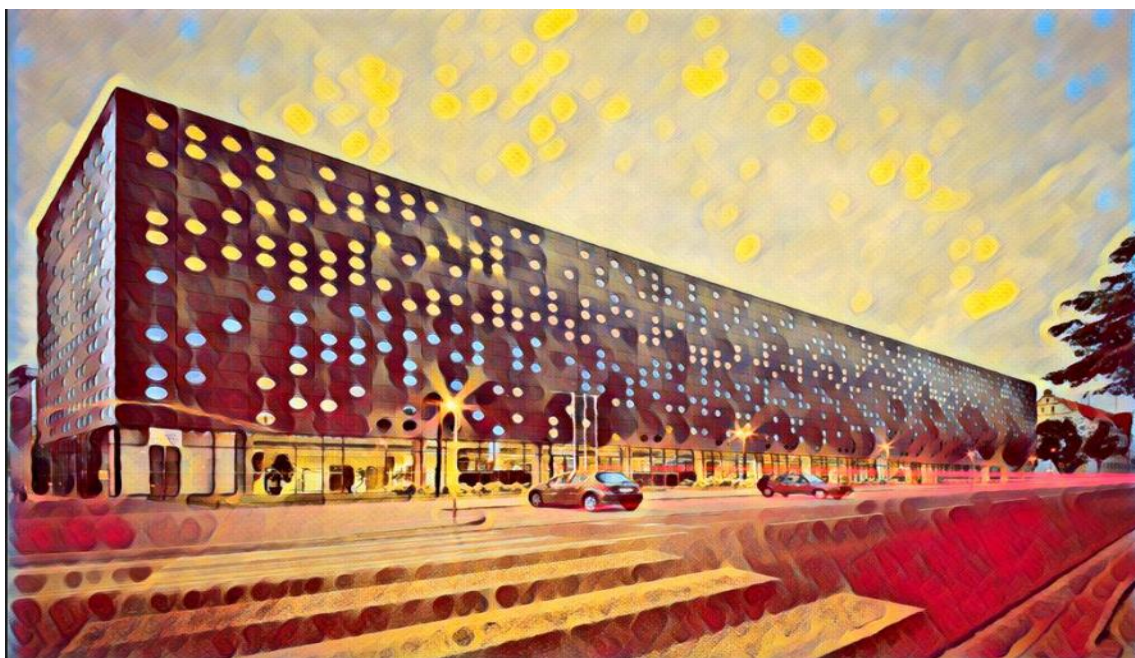


FIG 4

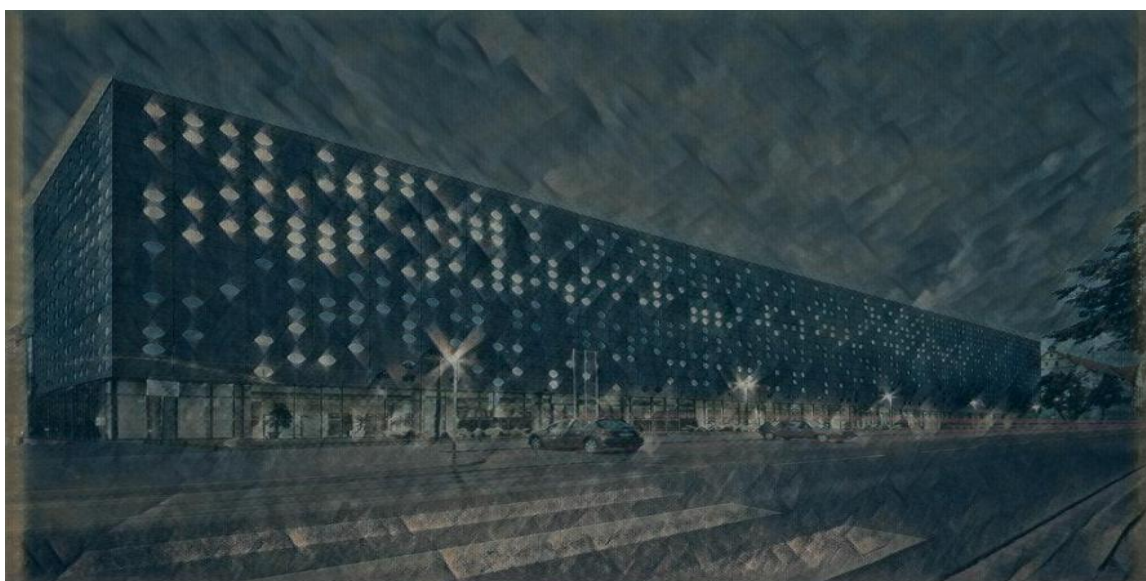


FIG 5

CONCLUSIONS

The project **NeuralCanvas – Real-Time AI Style Transfer** successfully demonstrates the use of deep learning techniques for artistic image transformation. The system combines the content of an input image with the artistic style of another image to generate visually appealing stylized outputs. By utilizing convolutional neural networks and a transformer network based on a MobileNet-inspired architecture, the system is able to perform style transfer efficiently while maintaining the structural details of the original image.

The implementation uses a pre-trained VGG16 network to extract feature representations and compute content and style losses during training. The use of Gram matrices allows the model to capture complex artistic patterns such as textures, colors, and brush strokes from the style image. Once trained, the transformer network can generate stylized images quickly, making the system suitable for real-time applications.

In addition to static image stylization, the project also supports real-time style transfer using webcam input. This demonstrates the capability of the system to process video frames efficiently and apply artistic transformations dynamically. The results show that the proposed approach produces high-quality stylized images while maintaining computational efficiency.

Overall, the **NeuralCanvas system highlights the potential of deep learning in creative applications such as digital art generation, image editing, and real-time visual effects**. The project demonstrates how modern neural networks can bridge the gap between technology and artistic creativity by enabling automated and efficient style transformation.

FUTURE SCOPE

The **NeuralCanvas – Real-Time AI Style Transfer** system demonstrates the potential of deep learning in artistic image transformation. However, there are several improvements and extensions that can be implemented in the future to enhance the capabilities of the system.

Firstly, the current system uses a separate trained model for each artistic style. In the future, the system can be improved by implementing **arbitrary style transfer techniques**, which would allow users to apply any style image without retraining the model. This would make the system more flexible and user-friendly.

Secondly, the project can be extended by supporting **multiple styles within a single model**. Users could dynamically select different styles while the system is running, enabling a more interactive and creative experience.

Another improvement would be the development of a **web-based or mobile application** for NeuralCanvas. Deploying the system as a web application would allow users to upload images and apply artistic styles directly through a browser interface.

The system can also be enhanced by improving the **quality and resolution of stylized images**. Using more advanced neural network architectures or higher resolution training data could produce more detailed artistic outputs.

In addition, the real-time webcam stylization feature can be expanded to support **video file stylization**, allowing users to transform entire videos into artistic styles.

Finally, integrating **GPU optimization and model compression techniques** can further improve the speed and efficiency of the system, making it suitable for deployment on mobile devices and edge computing platforms.

Overall, these improvements would make the **NeuralCanvas system more powerful, scalable, and accessible for real-world creative applications**.

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